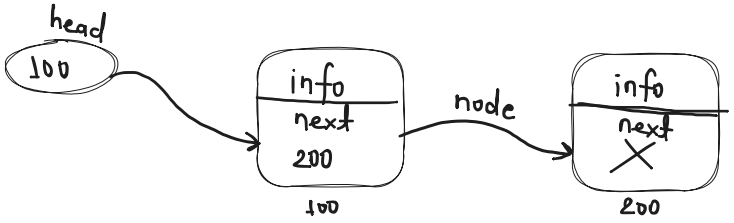
Linkedlist

A linkedlist is a linear data structure, in which elements are stored in separate object called nodes. Each node data and a reference(link) to another node.



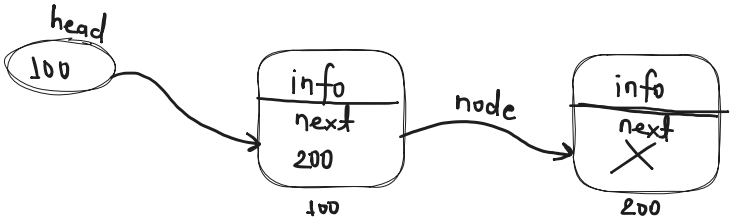
A linkedlist node contains :

- Data : The value stored in the node
- Next : A reference to the next node

--> Type of linkedlist:

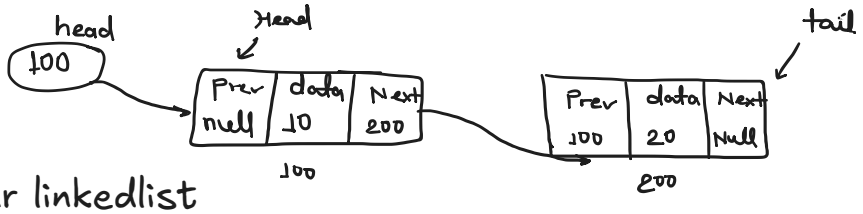
1. Single Linkedlist

Each node points to the next node only.



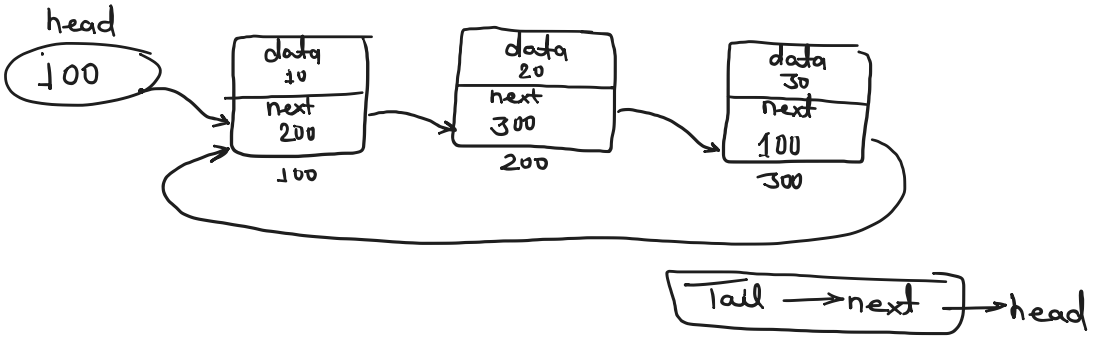
2. Doubly linkedlist

Each node points to both the previous and next nodes.



3. Circular linkedlist

The last node points back the first node instead of null.

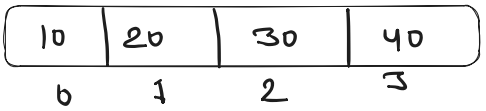


How does linkedlist overcome from the problem of array -

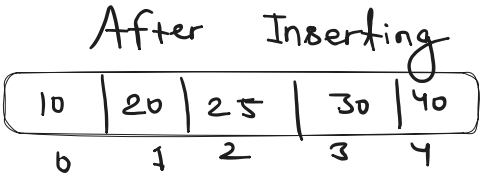
A linkedlist solve some of limitations of array specially when inserting or deleting element frequently.

Problem with array

Array Before Inserting

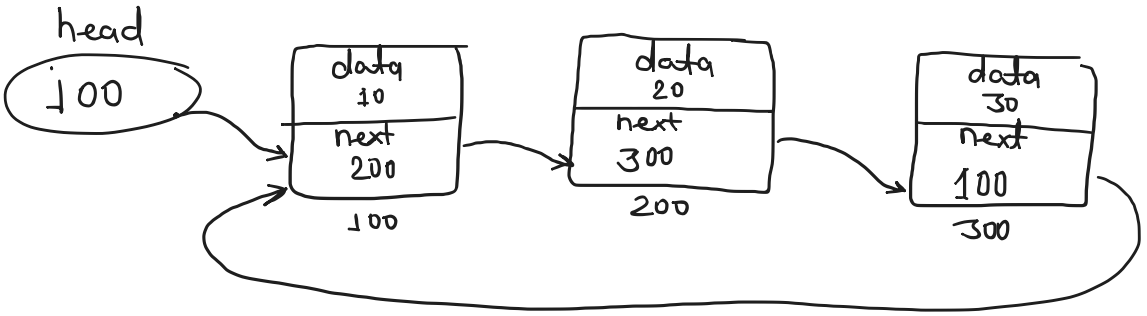


suppose, we want to insert 25 between 20 & 30.



we must shift 30 & 40 to the right in a commercial array. If the array is full, we may also needed to create a larger array and copy its element.

--> A basic Java array has fixed length.
--> Inserting the middle may required shifting element.



```
class Node{
    int data;
    Node next;

    Node(int data){
        this.data=data;
        this.next=null;
    }
}
class Main{
    public static void main(String arg[])
       {
        Node first = new Node(10);
        Node second = new Node(20);
        Node third = new Node(30);

    //connect the node
    first.next = second;
    second.next = third;

    //traverse the linkedlist
    Node current = first;
    while(current!=null)
        {
    System.out.println(current.data+"->");
    current = current.next;
        }
    System.out.println("null");
    }
        }
```